CS-736: Mathematical Imaging and Image Reconstruction

Assignment – X-Ray Computed Tomography

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Question 1: X-Ray Computed Tomography – Radon Transform

Objective

The objective of this problem is to implement the Radon transform of a 2D image by numerically approximating line integrals at multiple projection angles and offsets, and to analyze the effects of discretization and sampling choices on the resulting sinograms.

(a) X-ray Line Integration

Given an image $f(x, y)$ defined on a discrete grid, the Radon transform is defined as

$$\mathcal{R}f(t, \theta) = \int_{-\infty}^{\infty} f(x, y)\, \delta(x \cos\theta + y \sin\theta - t)\, dx\, dy.$$

Using a parametric representation of the line

$$x(s) = t\cos\theta - s\sin\theta,$$
$$y(s) = t\sin\theta + s\cos\theta,$$

the transform can be written as

$$\mathcal{R}f(t, \theta) = \int_{-\infty}^{\infty} f(x(s), y(s))\, ds.$$

Since the image is discrete, this integral is approximated using a summation:

$$\mathcal{R}f(t, \theta) \approx \sum_{k} f(x(s_k), y(s_k))\, \Delta s,$$

where Δs is the sampling step size along the ray.

Choice of Δs. The Radon transform is computed using $\Delta s = 0.5$, 1.0, and 3.0 pixel-width units. Smaller values provide better numerical accuracy at the expense of higher computational cost, while larger values lead to undersampling and approximation errors.

Interpolation scheme. Bilinear interpolation is used to evaluate image intensities at off-grid locations. This choice offers a good trade-off between accuracy and computational efficiency while preserving linearity of the transform. Pixels outside the image domain are assigned zero intensity.

(b) Discrete Radon Transform

Using the integration function from part (a), the Radon transform is computed for the discrete sets

$$t = -90, -85, \ldots, 85, 90, \quad \theta = 0, 5, 10, \ldots, 175.$$

For each pair (t, θ), numerical integration is performed along the corresponding line to obtain a discrete sinogram.

Placeholder: Radon transform (sinogram)
$\Delta s = 1.0$, $t \in [-90, 90]$, $\theta \in [0, 175]$

Figure 1: Discrete Radon transform of the Shepp–Logan phantom.

(c) Effect of Integration Step Size

Radon-transform images are computed for three different step sizes: $\Delta s = 0.5$, 1.0, and 3.0 pixel-width units.

1D Radon profiles. For $\theta = 0°$ and $\theta = 90°$, the 1D Radon profiles computed with $\Delta s = 0.5$ appear the smoothest, while those computed with $\Delta s = 3.0$ appear the roughest due to undersampling along the integration direction.

Placeholder: 1D Radon profiles
$\theta = 0°$ and $90°$ for different Δs

Figure 2: Comparison of 1D Radon-transform profiles for different integration step sizes.

2D Radon-transform images. The Radon-transform image corresponding to $\Delta s = 0.5$ is the smoothest, while the image corresponding to $\Delta s = 3.0$ shows visible artifacts and loss of detail.

Placeholder: Radon images for
$\Delta s = 0.5,\ 1.0,\ 3.0$

Figure 3: Effect of integration step size on the Radon-transform image.

(d) CT Scanner Parameter Design

In designing an X-ray CT scanner, smaller values of Δt and $\Delta\theta$ are preferred to ensure dense spatial and angular sampling. Smaller Δt improves spatial resolution, while smaller $\Delta\theta$ reduces angular undersampling artifacts. However, smaller values increase acquisition time, radiation dose, and storage requirements, leading to a trade-off between image quality and practical constraints.

(e) ART Reconstruction Considerations

Choice of pixel grid. The pixel size should be chosen small enough to capture anatomical details but large enough to avoid an excessively ill-conditioned reconstruction problem.

Effect of Δs. Choosing $\Delta s \gg 1$ pixel width leads to inaccurate ray–pixel interactions and degraded reconstructions, while choosing $\Delta s \ll 1$ pixel width improves accuracy but increases computational cost with diminishing returns.